

2025 Fall MAT2001 Midterm

October 20, 2025

Note: Only Q1, Q3, Q5, and Q8 are identical to the problems on the exam. The remaining questions are adapted from the homework assignments, which are very similar to the exam content. In particular, Q2 is harder than the one on the exam, while Q9 corresponds to the important Sturm Comparison Theorem, which was not covered in the lecture, though a similar question will also appear on the final exam

Q1. Solve the initial-value problem $t \frac{dy}{dt} = y + \sqrt{t^2 + y^2}$, $y(1) = 0$.

Q2. A tank contains 300 gallons of water and 100 gallons of pollutants. Fresh water is pumped into the tank at the rate of 2 gal/min, and the well-stirred mixture leaves at the same rate. How long does it take for the concentration of pollutants in the tank to decrease to 1/10 of its original value?

Q3. Consider the initial-value problem $y' = t^2 + y^2$, $y(0) = 0$. Suppose a solution exists in a neighborhood of $t = 0$. Prove that the solution is unique, without directly applying the existence and uniqueness theorem.

Q4. Compute the first three Picard iterates for the initial-value problem $y' = e^t + y^2$, $y(0) = 0$.

Q5. A small ball of mass m is thrown vertically upward from the ground with an initial velocity v_0 . The magnitude of air resistance is proportional to the square of the ball's velocity.

1. Establish the differential equation model of motion for the ball in the rising phase;
2. Find the relationship between velocity and height during the falling process;

3. Find the magnitude of the velocity when the ball falls back to the original point.

Q6. Find the general solution of the differential equation $y'' + 4y = t \sin 2t$.

Q7. Solve the initial-value problem

$$t^2 y'' - ty' - 2y = 0; \quad y(1) = 0, \quad y'(1) = 1$$

Q8. Suppose that y_1 and y_2 are a fundamental set of solutions on the interval $-\infty < t < \infty$. Show that there is one and only one zero of y_1 between consecutive zeros of y_2 .

Q9. (i) Given the following second-order linear homogeneous ordinary differential equations:

$$y'' + r(x)y = 0, \quad z'' + q(x)z = 0$$

where $r(x) > q(x)$ holds for all x in the domain of definition, and y, z are non-trivial solutions of the respective equations. Show that between any two consecutive zeros of z , there exists at least one zero of y .

(ii) Consider the following second-order linear homogeneous ordinary differential equation:

$$y'' + \left(1 + \frac{\sin^2 x}{1 + x^2}\right)y = 0$$

Show that, the function $y(x)$ has infinitely many zeros. Furthermore, when $x \rightarrow \infty$, the distance between any two consecutive zeros of $y(x)$ tends to π .

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